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Guest editors: Prof. Filipe Alvelos, Prof. Jaime Carrasco Barra, Prof. Miguel Constantino, and Prof. Yu Wei

Dear Professors Alvelos, Carrasco Barra, Constantino, and Wei,

We are pleased to submit our manuscript “The Cycle-Constrained Aerial Suppression Base and Water Point Location Problem: Joint Siting under Uncertainty with an Application to Cundinamarca, Colombia” to the International Transactions in Operational Research. As requested in the call for papers, we indicate that this submission is intended for the special issue on Operations Research in Wildfire Management.

The paper sits squarely within the call’s scope, in particular the listed topics of resource location-allocation in wildfire preparedness, combined problems, budgeting, and suppression resource operations, approached with stochastic programming and matheuristics. It introduces, to our knowledge, the first model that sites aerial suppression bases and water refill points jointly, with aircraft productivity endogenous to the base to fire to water refill cycle, as a two-stage mean-risk stochastic program based on conditional value-at-risk (CVaR); a single budget covering bases, aircraft, and water infrastructure makes the budget split endogenous. An exact linearization and a fix-and-optimize matheuristic with provably exact reduced subproblems make the full candidate catalog tractable. A fully open-data, replicable case study for Cundinamarca, Colombia, is built around the country’s real Sikorsky S-70 Firehawk acquisition and includes a sample average approximation replication analysis with an optimality-gap estimate and an operational reading of the optimal plan in real places, yielding policy-relevant structural findings on when sequential planning suffices and on what limits containment.

The manuscript is original, unpublished, and not under consideration elsewhere. The research received no external funding and the authors declare no competing interests. All data sources are open and every experiment is reproducible from the companion repository, available to reviewers upon request.

Sincerely,

Jaime Enrique Pesca Santos (corresponding author, jaime.pesca@uexternado.edu.co)
Facultad de Administración de Empresas, Universidad Externado de Colombia, Bogotá, Colombia

Kiana Hikaru Ysa Morla (kiana.ysa@utec.edu.pe)
Universidad de Ingeniería y Tecnología (UTEC), Lima, Peru